



MINING DEALER BEST PRACTICE

Drilling Deck Platform MD6420

**Maintenance and
Repair**

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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August 2022
1.10-220824-1

1.0 Introduction

Maintenance and repair work on a drilling rig can be risky and time-consuming. Working over the trench of the drilling platform while add proper name for risky activity here “tensioning pulldown” is just one key example of such an activity. It is important for the mechanic to have a safe footing while conducting this maintenance activity so he/she can concentrate on the activity and not have to worry about falling into the trench.

The typical solution used by customers is placing scrap wood over the “trench” which also present additional risk and time challenges - proper size, proper thickness, good condition-not rotted, being secure, searching for the scrap wood each time maintenance is needed, handling the improvised platform, etc. The photos in Figure 1 show this situation.

The customer and maintenance personnel asked Sotreq to come up with a solution to support efficient and effective, risk and time minimizing maintenance on the drilling deck.



Figure 1: Scenario before Best Practice implementation

2.0 Best Practice Description

This Best Practice describes Sotreq’s solution of a Drilling Deck Maintenance Platform that provides and supports efficient, effective, and safe maintenance work behind the mast when the mast is lowered.

The development and implementation of the platform is the result of collaboration between engineering teams and product support from SOTREQ and the customer's drilling team at the mine.

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While this solution for this site was for the MD6420, it can easily be developed for other drill models.

Some notes:

Each machine will have its own platform.

- When not in use, the platforms should be stored in a known/staged location so that they are always readily available.
- To avoid loss or damage to the equipment, supports were installed on the handrails on the left and right sides of the platform to allow lifting and moving to position it in a safe manner.

3.0 Implementation Steps

Implementation of this best practice solution is broken into two key steps

1. Design and Build the Platform
2. Assembly

Step 1: Design & Build Platform

- Interview the mechanics and analyze the problem locally
- Review potential solutions / Choose the best solution
- Measure work area (gap/trench)
- Determine best material to make the platform out of
- Design in Autocad (see diagrams below)
- Build main platform
- Check for Proper Fit / Adjust as necessary.

Step 2: Assembly

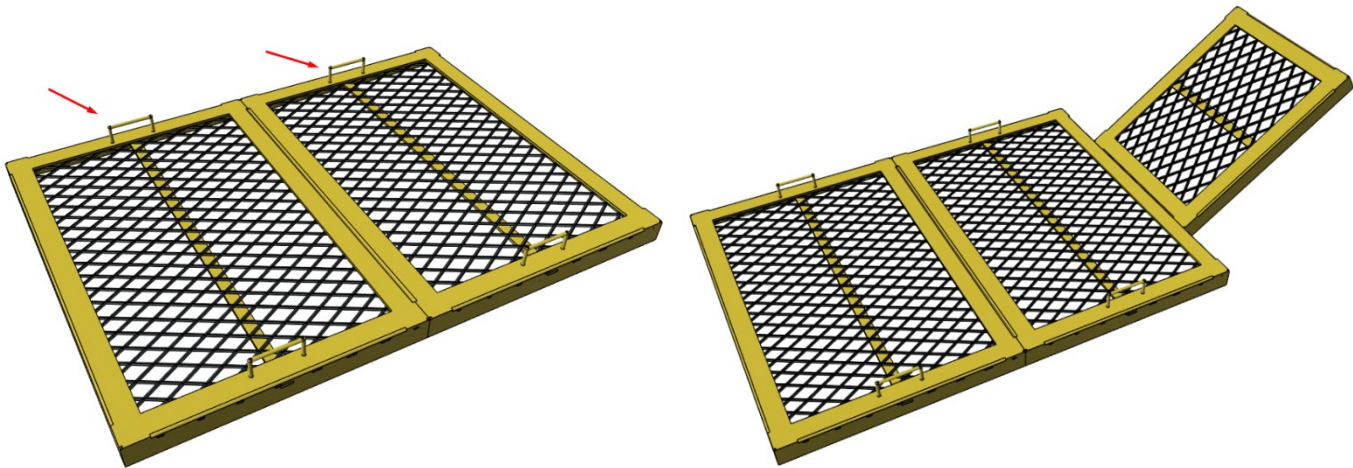
- Add handles/supports (allowing it to be mobile) for handling during maintenance.



Protection on equipment

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Targeted protection

- Add support for fixing to the left side handrail in line with the cab.



Figure 2: Platform support oriented location

- Test platform in place.

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4.0 Benefits

Key Benefits of implementing this best practice include the following:

- Improved Safety in working area
- Improved Efficiency and Effectiveness (confidence in stability of working area)
- Easy installation and replication for other machine models

BP in Action: The platform is being used daily on demand. In the images below, there is evidence of its use when performing a service with the mast lowered, which proves the greater safety and convenience for mechanics.

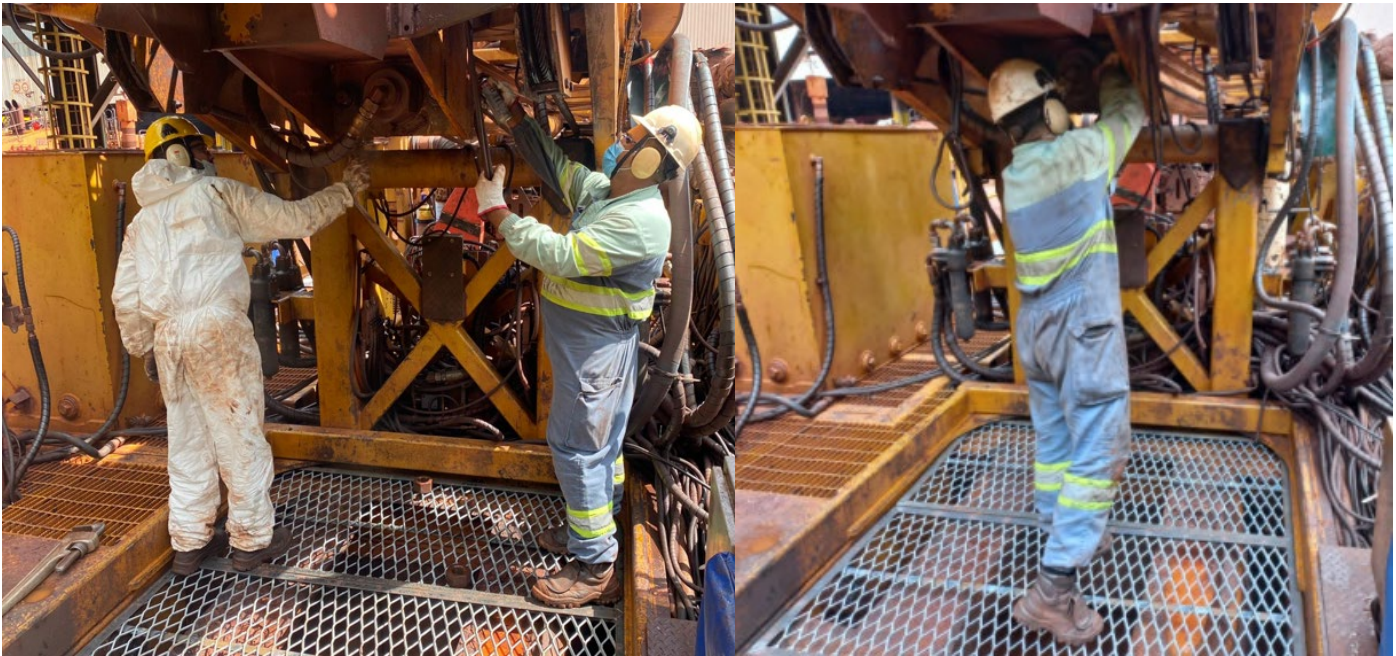


Figure 3: Evidence of platform use

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5.0 Resources Required

1020 steel for grid production.

“AutoCad” software for producing the graphic expression of the project

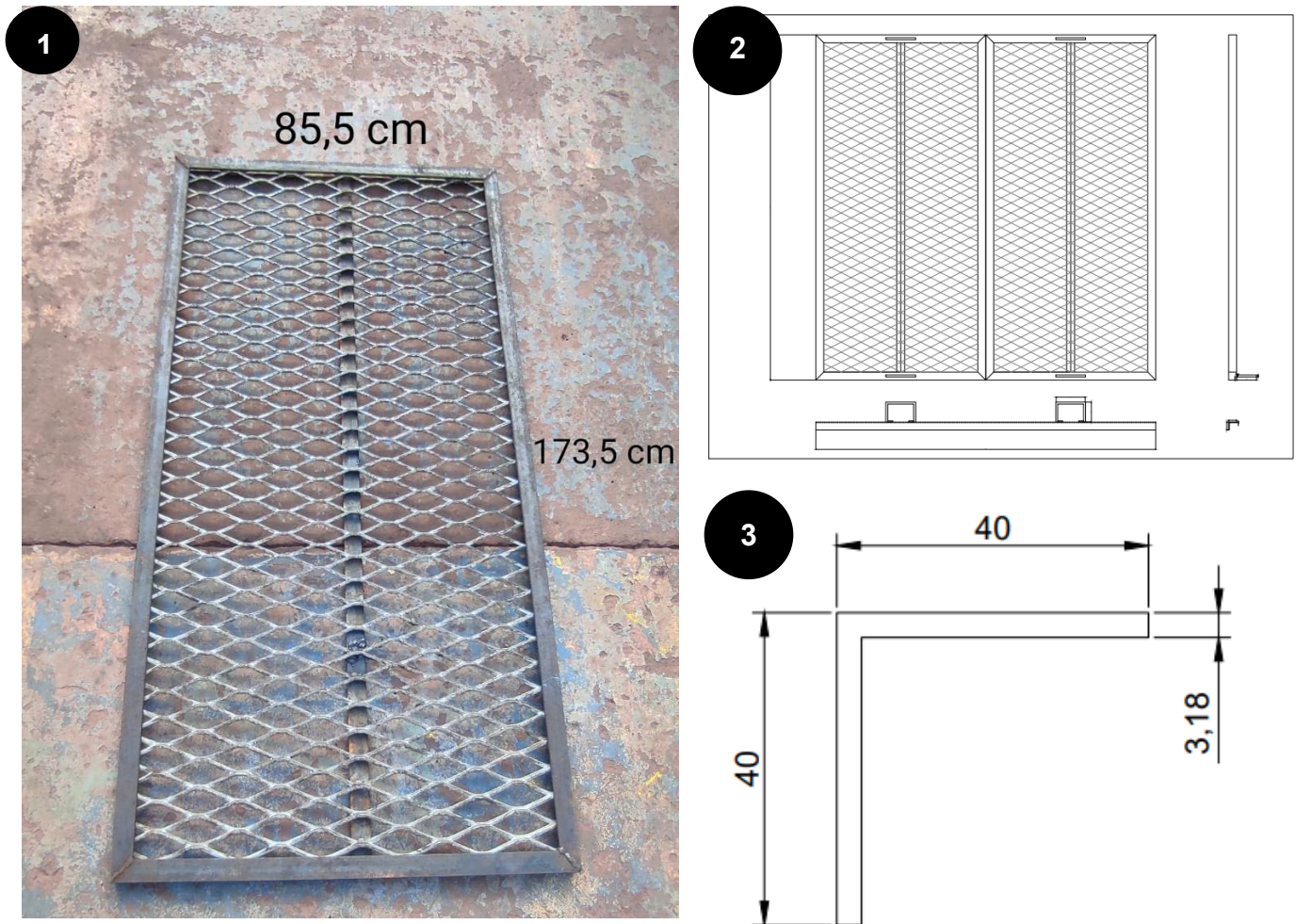


Figure 4: Measures used to build the platform

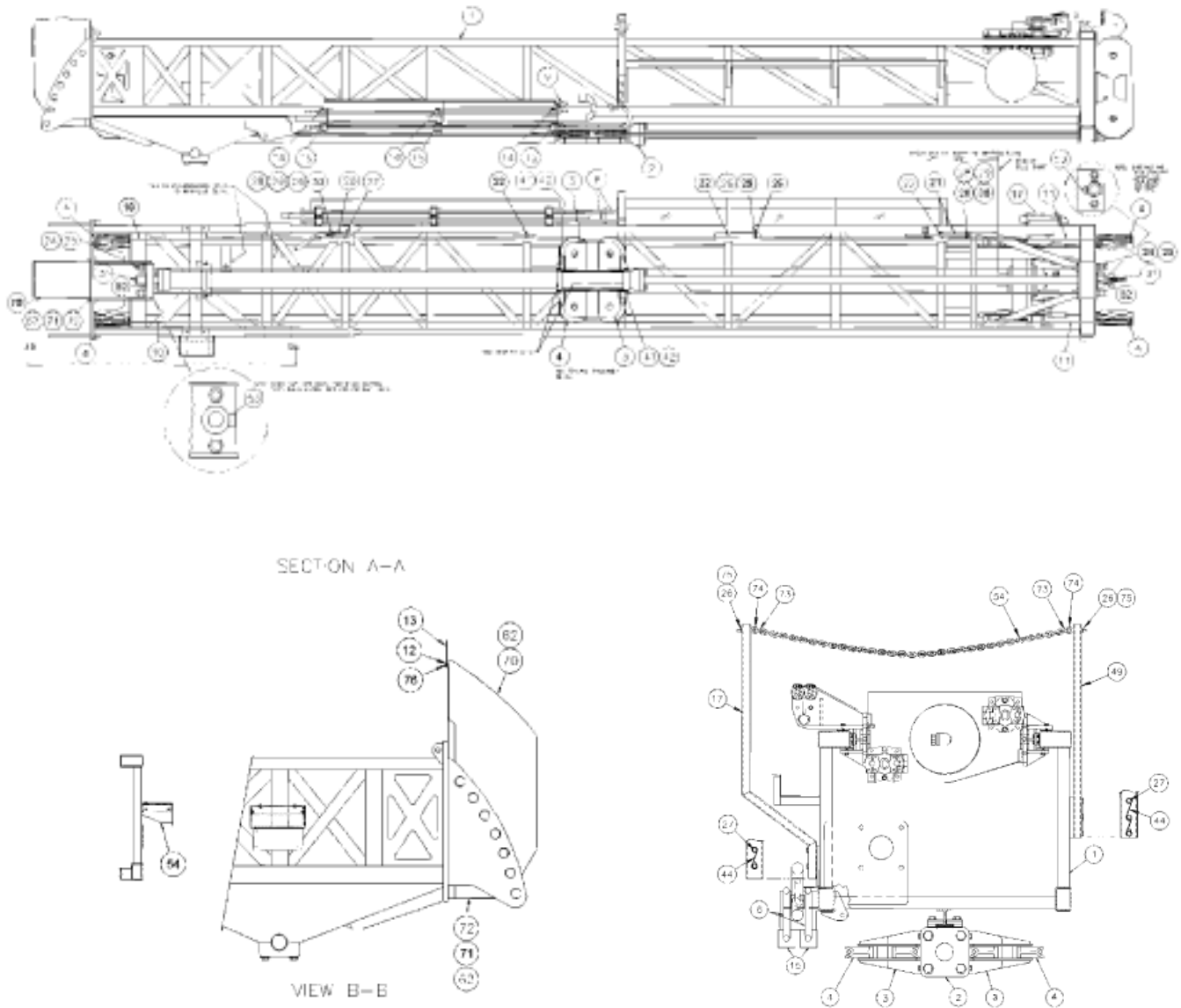
Espessura		Espessura		Peso Teórico Kg/m ²
pol.	mm	B mm	A mm	
1/8"	3,00	100	50	2,80
3/16"	4,75	100	50	5,60
1/4"	6,35	100	50	8,40

Manpower to Design: ~4 hours

Manpower to Build: ~8 hours

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6.0 Supporting Attachments / References



7.0 Related Best Practices

N/A

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8.0 Acknowledgements

We would like to thank everyone involved in this improvement, as well as the customer's trust in our suggestion.

Raissa Maués Nascimento
Fleet Performance Engineer
Sotreq S/A

Gladyson Silva
Technical Consultant
Sotreq S/A

Thiago Dillon
Technical Analyst
Sotreq S/A:

Glauber Nogueira
Predictive Technician
Sotreq S/A

Fernando Melo
Technical Mechanic
Sotreq S/A

Felisberto Dias
Technical Mechanic
Sotreq S/A

Elias Nogueira
Technical Mechanic
Sotreq S/A

Francisco Lopes
Technical Mechanic
Sotreq S/A

Josinei Ataide Valentim
Fleet Performance Engineer
Sotreq S/A

Isaac Pedrosa
Engineering Consultant
Sotreq S/A

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